

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2023

ME 2115

(Fluid Mechanics)

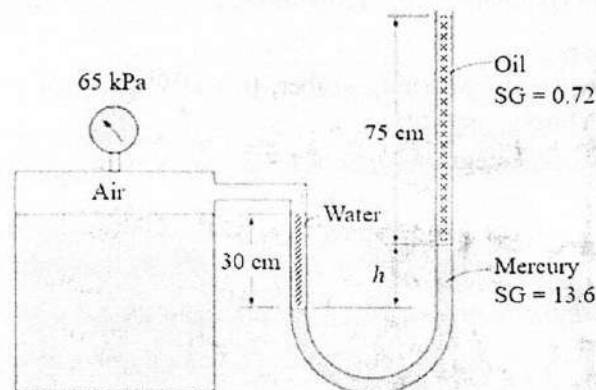
Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Explain the Newton's law of viscosity. 09
- 1(b). What is meant by capillary effect? Show that the capillary rise $h = \frac{2\sigma_s}{\rho g R} \cos\phi$, where the symbols have their as usual meanings. 10
- 1(c). What is cavitation? What is its importance? 06
- 1(d). A 1.2 mm diameter tube is inserted into an unknown liquid whose density is 980 kg/m^3 , and it is observed that the liquid rises 6 mm in the tube, making a constant angle of 15° . Determine the surface tension of the liquid. 10
- 2(a). Describe various pressure measurement devices. 10
- 2(b). Derive expressions for the resultant force and center of pressure in a submerged inclined surface. 15
- 2(c). The gage pressure of the air in the tank shown in figure is measured to be 65 kPa. Determine the differential height h of the mercury column. 10



- 3(a). What is meant by meta centric height? Explain the stability of a floating body. 10
- 3(b). Show that the buoyant force acting on a plate equal to to the weight of liquid displaced by the plate. 08
- 3(c). Distinguish between Eulerian and Lagrangian method of fluid flow. 07
- 3(d). A uniform body of size $3 \text{ m} \times 2 \text{ m} \times 1 \text{ m}$ float in water. What is the weight of the body if the depth of immersion is 0.8 m. 10

4(a).	Derive Reynolds transport theorem.	18
4(b).	What is conservation law of mass? Derive continuity equation applying law of conservation of mass.	12
4(c).	Distinguish between path line and streak line.	05

SECTION – B

5(a).	Derive Euler's equation of motion.	10
5(b).	What is the hydraulic grade line? How does it differ from the energy grade line? Under what conditions do both lines coincide with the free surface of a liquid?	06
5(c).	What are the limitations of Bernoulli's equation?	07
5(d).	A 15 hp (shaft) pump is used to raise water to a 55 m higher elevation. If the mechanical efficiency of the pump is 85%, determine the maximum volume flow rate of water.	12
6(a).	What is the function of a Pitot-tube?	05
6(b).	Differentiate between notch and weir.	05
6(c).	Prove that for right-angled V-notch the discharge is $1.42H^{5/2}$ considering $C_d = 0.6$.	15
6(d).	A pipeline carrying oil of SG 0.87 change in diameter from 200 mm(A) to 500 mm(B) which is 4 m at a higher level. If the pressure at A and B are 9.81 N/m^2 and 5.886 N/m^2 , respectively and discharge is 200 L/s, determine the loss of head and direction of flow.	10
7(a).	What is boundary layer sub-layer? Explain the phenomena of flow separation.	10
7(b).	Define the following: (i) Froude's number, (ii) Mach's number, (iii) Displacement thickness, (iv) Momentum thickness and (v) Energy thickness.	10
7(c).	Deduce momentum integral boundary layer equation with non-zero pressure gradient.	15
8(a).	Explain the effect of pressure gradient on boundary layer separation.	10
8(b).	What is skin friction and pressure drag? Explain the situation of their impact.	10
8(c).	A thin plate is moving in still atmosphere at a velocity of 5 m/s. The length of the plate is 0.6 m and width 0.5 m. Calculate (i) boundary layer thickness at the end of the plate and (ii) drag force on the side of the plate. Take $\rho_a = 1.24 \text{ kg/m}^3$ and $\nu = 0.15 \text{ stokes}$.	15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2023

Math 2113

(Linear Algebra and Vector Analysis)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). For what values of “ λ ” the following system of linear equations: 12

$$x + y + z = 1$$

$$x + 2y + 4z = \lambda$$

$$x + 4y + 10z = \lambda^2$$

Provides i) unique solution, ii) no solution and iii) infinitely many solution?

- 1(b). If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ then find A^{-1} by using adjoint process also find $A^{-1}B$. 13

- 1(c). Find a, b, c such that the matrix $\begin{pmatrix} a & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ b & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ c & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$ is orthogonal. 10

- 2(a). Solve the following matrix equation by using Gaussian elimination method: 13

$$AX = B \text{ Where } A = \begin{pmatrix} 1 & 2 & -3 \\ 1 & 3 & 1 \\ 2 & 5 & -4 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} 4 \\ 11 \\ 13 \end{pmatrix}$$

- 2(b). Reduce the following matrix to its row echelon form and finally normal form 12

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & -3 & -6 \\ 3 & -3 & 1 & 2 \end{bmatrix}$$

- 2(c). Define subspace of a vector space. 10
Verify whether or not $T = \{(a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0\}$ is a subspace of the vector space $\mathbb{R}^4$.

- 3(a). Test the linear dependence of the set $\{(1, -2, 1), (2, 1, -1), (7, -4, 1)\}$ 07

- 3(b). Consider the basis $s = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$, where $v_1 = (1, 1, 2)$, $v_2 = (1, 1, 0)$, and $v_3 = (1, 0, 0)$, and $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be linear operator such that $T(v_1) = (2, -1, 4)$, $T(v_2) = (3, 0, 1)$, $T(v_3) = (-1, 5, 1)$. Find a formula for $T(x_1, x_2, x_3)$ and use that formula to find $T(2, 4, -1)$. 14

- 3(c). Let U and W be subspaces of $\mathbb{R}^5$. $U = L\{(1, 3, -2, 2, 3), (1, 4, -3, 4, 2), (2, 3, -1, -2, 9)\}$ 14
 $W = L\{(1, 3, 0, 2, 1), (1, 5, -6, 6, 3), (2, 5, 3, 2, 1)\}$. Find the basis and dimension of i) U and ii) W

- 4(a). Find a basis for the row space consisting entirely of row vectors from the following matrix 12
- $$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$
- 4(b). Find all the eigen values and corresponding eigen vector corresponding to the largest eigen value of the following matrix 14
- $$\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$
- 4(c). For which value of λ will be the vector $v = (1, \lambda, 5)$ in $\mathbb{R}^3$ is a linear combination of the vectors $v_1 = (1, -3, 2)$, $v_2 = (2, -1, 1)$ 09

SECTION – B

- 5(a). Sketch the space curve $x = 3 \cos t$, $y = 3 \sin t$, $z = t$ and find i) the unit tangent , ii) the principal normal, curvature, iii) the binormal, torsion 15
- 5(b). Define vector-valued function with an example. A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$, where t is the time. Find the magnitudes of the velocity and acceleration at $t = 0$. 10
- 5(c). Evaluate : ∇r^n and $\nabla \varphi$ where $\varphi = \ln|\vec{r}|$ 10
- 6(a). Find the directional derivative of the function $\varphi(x, y, z) = \frac{y}{x+z}$ at the point $P(2, 1, -1)$ in the direction towards the point $Q(-1, 2, 0)$ 11
- 6(b). Find an equation for the tangent plane to the hyperboloid given by $z^2 - 2x^2 - 2y^2 = 12$ at the point $(1, -1, 4)$. Find also normal line to the surface at the given point in symmetrical form. 12
- 6(c). Define solenoidal vector. Prove that the vector $\vec{F} = (2x^2 + 8xy^2z)\hat{i} + (3x^3y - 3xy)\hat{j} - (4y^2z^2 + 2x^3z)\hat{k}$ is not solenoidal but $\vec{G} = xyz^2$. $\vec{F}$ is solenoidal. 12
- 7(a). Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is a conservative force field. Find the scalar potential for $\vec{F}$, Also Find the work done in moving an object in this field from $(0, 1, -1)$ to $(\frac{\pi}{2}, -1, 2)$. 15
- 7(b). Evaluate $\iint_S \vec{F} \cdot \vec{n} \, ds$ using the law of projection, where $\vec{F} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the first octant bounded by the part of the plane $2x + y + 2z = 6$. 14
- 7(c). Define jacobian of two functions $u = u(x, y)$ and $v = v(x, y)$. Hence find $J(u, v)$ where $u = 2x + 5$, $v = x - y$. 06
- 8(a). If R is the region bounded by the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$ above the xy - plane. find the value of $\iint_R \sqrt{(x^2 + y^2)} \, dx dy$, using the Jacobian Transformation. 10
- 8(b). Evaluate $\oint_C (xy + y^2)dx + x^2 dy$ by using Green's Theorem, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. 10
- 8(c). Verify the divergence theorem for $\vec{A} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by $x^2 + y^2 = 4$, $z = 0$, and $z = 3$. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination, 2023

EE 2113

(Electrical Machines)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define i) Magnetomotive force, ii) Coercive force, iii) Reluctance, iv) Magnetic flux, v) Curie Temperature. 10
- 1(b). Justify the statement that- “Magnetic circuit is analogous to the electrical circuit”. 10
- 1(c). why does falling copper ring slow down as it approaches to a permanent magnet? Explain. 05
- 1(d). State and explain Faradays law of electromagnetic theory. 10
- 2(a). Define critical resistance. Explain the voltage build up process of a DC shunt generator. Also, mention the factors affecting voltage build up process. 13
- 2(b). Classify different types of generator. Draw the three characteristic curves with brief explanation of DC shunt and series generator. 12
- 2(c). A DC short shunt lap wound 4 pole generator has armature resistance, shunt, and series resistances are 0.05 ohm, 100 ohm, and 0.02 ohm, respectively. If the generator supplies 220 V at 50A to the load, find the armature current and generated emf. 10
(Given brush contact drop =1 V brush)
- 3(a). How does a DC motor works? Draw and explain the speed -torque characteristic curve of the DC shunt motor. 13
- 3(b). Suppose you want to build a sun tracking device for solar panel, which type of DC motor is required for that? Draw the characteristic curves of that motor and explain the reasons. 12
- 3(c). A DC shunt motor gives 3000 rpm of rotation by taking 10 kW power at 100 A load. Given that, the motor is wave wound with four pole, producing magnetic flux of 5 mWb. If the motor armature has 200 number of conductors, then 10
i) What will be the back emf developed in the motor?,
ii) If the motor is lap wound, then what will be number of armature conductor required to produce the same mechanical rotation?
- 4(a). Justify the statement that “A DC generator can be run as a DC motor if supply is given to the armature”. 10
- 4(b). Why DC motor needs starter? Describe the working principle of a 3-point starter with necessary schematics. 12
- 4(c). A belt-driven DC generator takes 100 kW at 200 A from a busbar. But when the belt breaks, it takes 10 kW at 50 A from the bus bar. If the DC machine has 0.05 Ω of armature resistance, 100 Ω of shunt field resistance and rotated at 500 rpm, what will be the generated emf and when the belt breaks, what will be the new speed of the machine? 13

SECTION – B

- 5(a). Explain the working principle and construction of a transformer. 10
- 5(b). What are the main purposes behind conducting short-circuit and open-circuit tests on a transformer? Explain with proper connection diagram. 10
- 5(c). The equivalent circuit impedances of a 20 KVA, 800/240 V, 60 Hz transformer are to be determined. The open-circuit test was performed on the secondary side of the transformer (to reduce the maximum voltage to be measured) and the short-circuit test were performed on the primary side of the transformer (to reduce the maximum current to be measured) the following data were taken: 15

Open circuit test	Short circuit test
$V_{oc} = 240 \text{ V}$	$V_{sc} = 483 \text{ V}$
$I_{oc} = 7.133 \text{ A}$	$I_{sc} = 7.133 \text{ A}$
$P_{oc} = 400 \text{ W}$	$P_{sc} = 240 \text{ W}$

Find the impedances of the approximate equivalent circuit referred to the primary side, and draw the circuit.

- 6(a). Define: i) pole pitch, and ii) spread-factor. 06
- 6(b). What are the different types of losses occurred in an alternator? Explain using power-flow diagram. 09
- 6(c). Explain the conditions of parallel operation of alternators. 10
- 6(d). The following test results are obtained from a 3 phase, 6600-KVA, 6600 V, star connected, 2 pole, 50 Hz turbo-alternator: 10
With a field current of 125 A, the open-circuit voltage is 8,000 V at the rated speed; with the same field current and rated speed, the short-circuit current is 800A. At the rated full-load, the resistance drop is 3%. Find the regulation of the alternator on full-load and at a power factor of 0.8 lagging.
- 7(a). Determine the condition for maximum torque of an induction motor under running condition. 12
- 7(b). Is single –phase induction motor self-starting? Explain with proper schematics. 13
- 7(c). A 3-phase, slip ring, induction motor with star-connected rotor has an induced emf of 120 V between slip rings at stand still with normal voltage applied to stator. The rotor winding has a resistance per phase of 0.3 ohm and standstill leakage reactance per phase of 1.5 ohm. Calculate: i) rotor current/phase when running short-circuited with 4% slip and ii) the slip and rotor current per phase when rotor is developing maximum torque. 10
- 8(a). Explain voltage regulation of an alternator. 05
- 8(b). What is all day efficiency of a transformer? What is its significance? 10
- 8(c). Draw the equivalent circuit of an induction motor. 05
- 8(d). Find the all-day efficiency of 500 KVA distribution transformer whose copper loss and iron loss at full load are 4.5 kW and 3.5 kW respectively. During a day of 24 hours, it is loaded given below: 15

No of hours	Loading in kW	Power factor
6	400	0.80
10	300	0.75
4	100	0.80
4	0	-

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 1st Term Examination 2023

ME 2113

(Statics and Solid Mechanics)

Time: 3 Hours.

Full Marks: 210

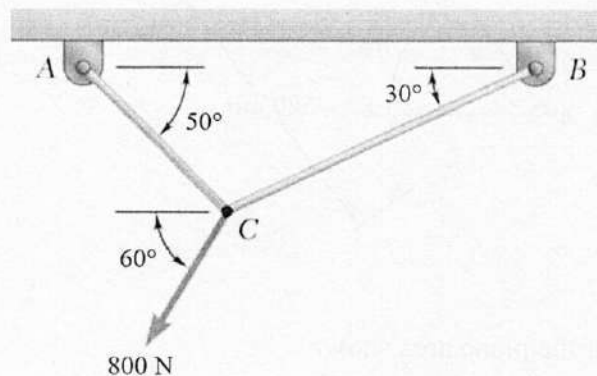
N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

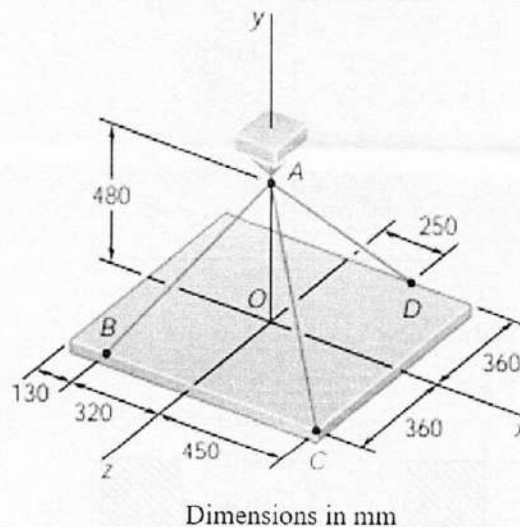
iii) Assume reasonable data if any missing.

SECTION – A

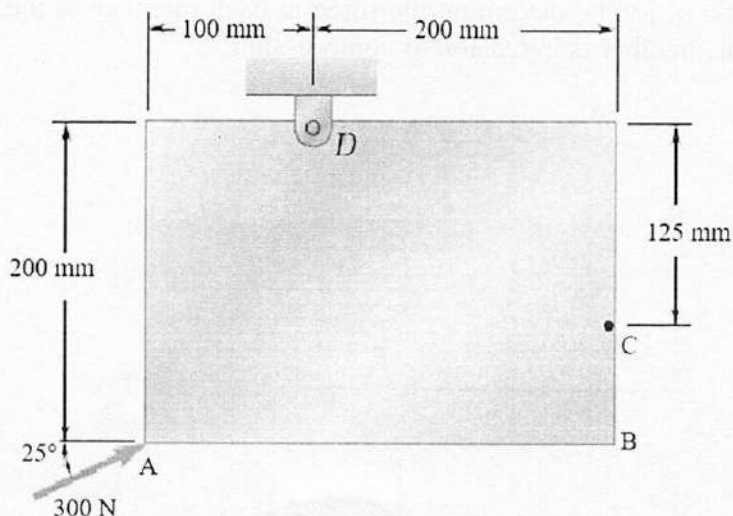
- 1(a) Two cables are tied together at C and are loaded as shown. Determine the tension in cable AC and BC. 17



- 1(b) A rectangular plate is supported by three cables as shown. Knowing that the tension in cable AC is 60 N, determine the weight of the plate. 18

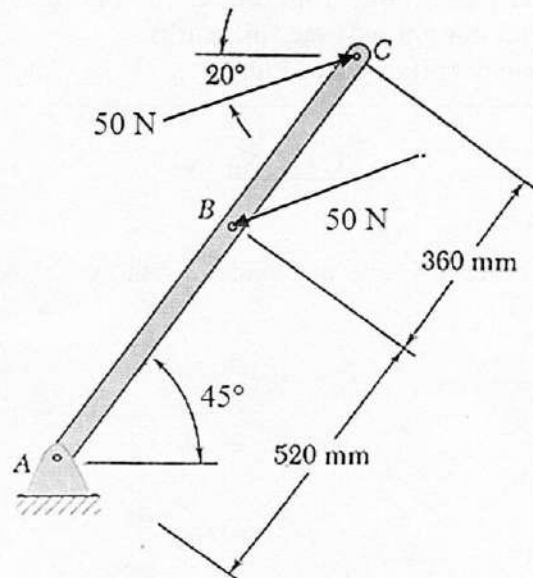


- 2(a) A 300 N force is applied at A as shown. Determine (i) the moment of the 300 N force about D, (ii) the magnitude and sense of the horizontal force applied at C that creates the same moment about D, (iii) the smallest force applied at C that creates the same moment about D. 18



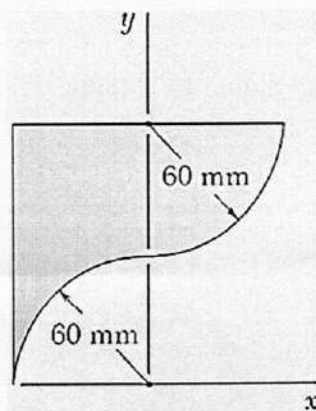
- 2(b) Two parallel 50 N forces are applied to a lever as shown. Determine the moment of the couple formed by the two forces.

17



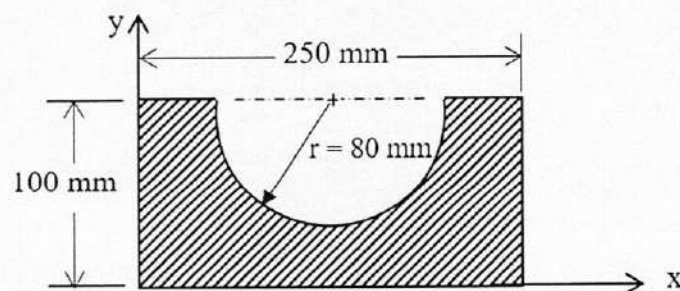
- 3(a) Locate the centroid of the plane area shown.

17



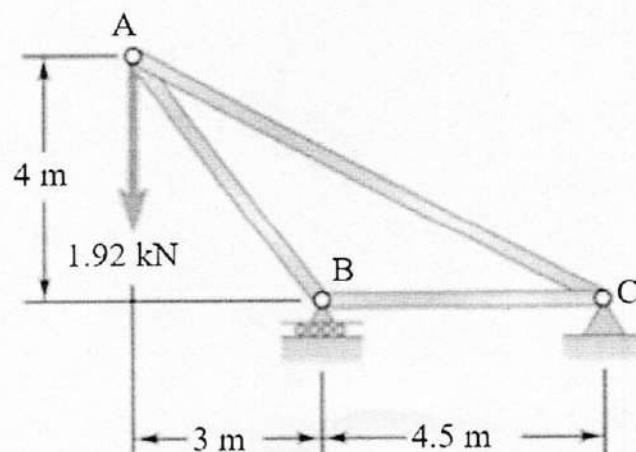
- 3(b) Determine the moment of inertia with respect to the x-axis.

18



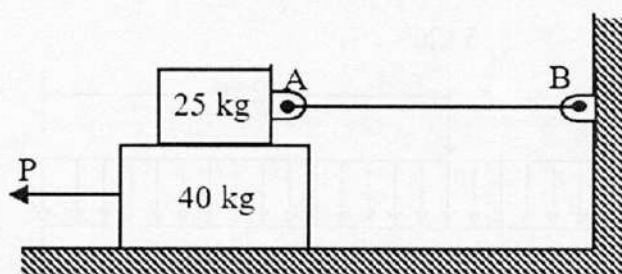
- 4(a) Using the method of joints, determine the force in each member of the truss shown. State whether the member is in tension or compression.

18



- 4(b) The coefficients of friction are $\mu_s = 0.30$ and $\mu_k = 0.25$ between all surfaces of contact. Determine the smallest force P required to start the 40 kg block moving if cable AB is (i) attached as shown, (ii) removed.

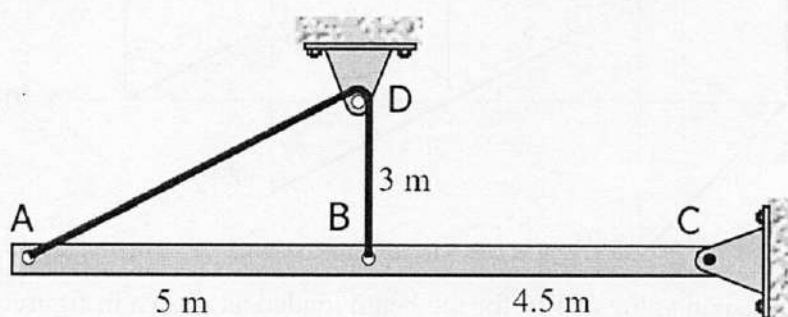
17



SECTION - B

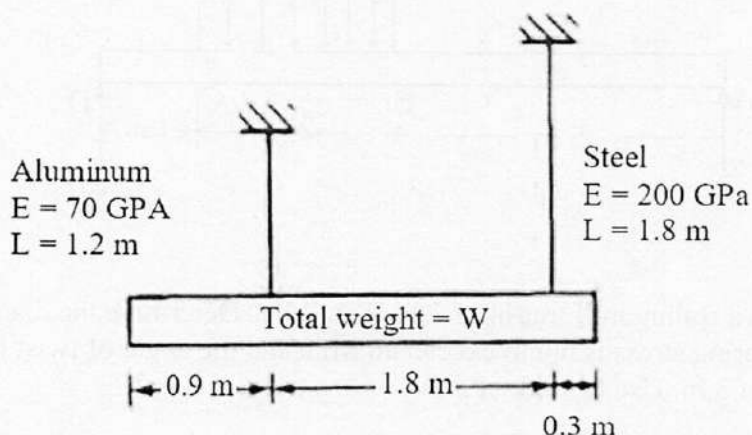
- 5(a) The homogeneous bar shown in figure is supported by a smooth pin at C and a cable that runs from A to B around the smooth peg at D. Find the stress in the cable if its diameter is 15 mm and the bar weighs 3000 kg. What would be the factor of safety if the allowable stress in the cable is 200 MPa?

17



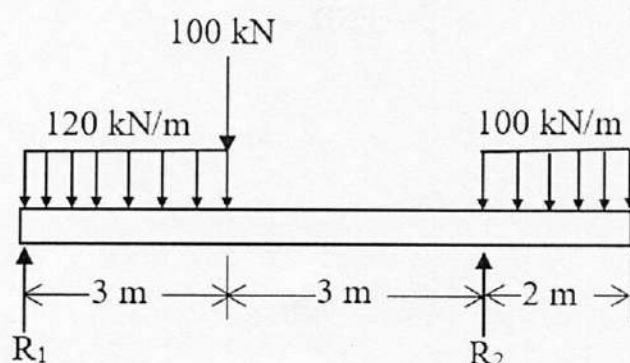
- 5(b) A uniform concrete slab of total weight W is to be attached as shown in figure to two rods whose lower ends are on the same level. Determine the ratio of the areas of the rods so that the slab will remain level.

18



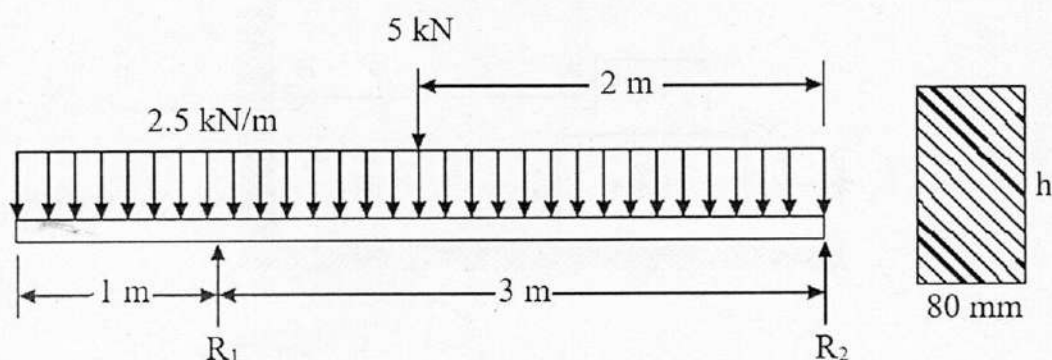
- 6(a) Draw the shear force and bending moment diagrams for the beam loaded as shown in figure.

17



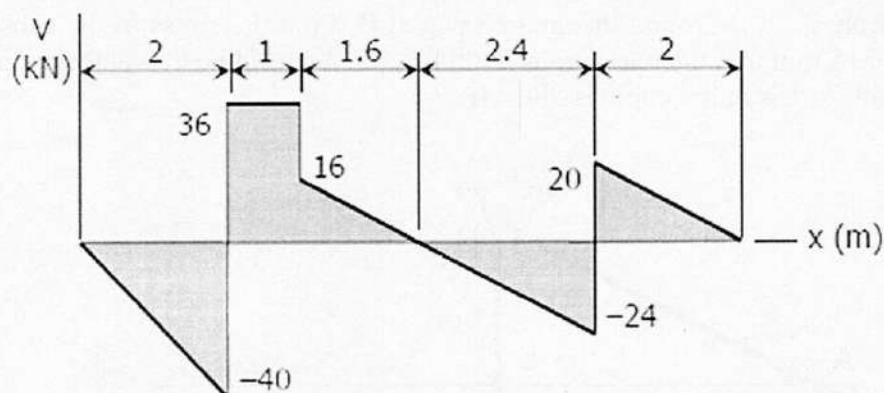
- 6(b) Determine the minimum height h of the beam shown in figure if the flexural stress is not to exceed 20 MPa.

18



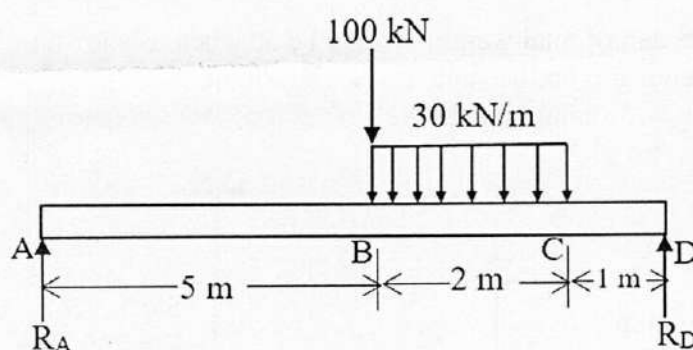
- 7(a) Draw moment and load diagram corresponding to the given shear diagram in figure.

15



- 7(b) Compute the midspan value of $EI\delta$ for the beam loaded as shown in figure.

20



- 8(a) A solid shaft in a rolling mill transmits 20 kW at 2 Hz. Determine the diameter of the shaft if the shearing stress is not to exceed 40 MPa and the angle of twist is limited to 6° in a length of 3 m. Use $G = 83 \text{ GPa}$.
- 8(b) Select the lightest W shape that will act as a column 10 m long with hinged ends and support an axial load of 300 kN with a factor of safety of 2.5. Assume that the proportional limit is 200 MPa and $E = 200 \text{ GPa}$.

17

18

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY**Department of Energy Science and Engineering**

B. Sc. Engineering 2nd Year 1st Term Examination, 2023

CSE 2113

(Computer Programming)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). The equation for extraterrestrial radiation is given by the following expression: 10

$$G_{on} = G_{sc} \left(1 + 0.033 \cos \frac{360x}{365} \right)$$

Another more correct equation is given by the following expression:

$G_{on} = G_{sc} (1.000110 + 0.034221 \cos B + 0.001280 \sin B + 0.000719 \cos 2B + 0.000077 \sin 2B)$, where value of G_{sc} is 1367 W/m^2 , n is the number of day and B is given by,

$$B = (n - 1) \frac{360}{365}$$

Now, write MATLAB code that calculates G_{on} using both of the equation for 365 days.

- 1(b). Here are given two polynomial equations: 14

$$P_1(x) = 3x^6 + 15x^5 - 10x^3 - 3x^2 + 15x - 40 \text{ and } P_2(x) = 3x^3 - 2x - 6$$

- Add, subtract, multiply and divide the given polynomials using MATLAB code.
- Calculate $P_1(3)$ and $P_2(6)$ using MATLAB code.
- Find roots of the given polynomials using MATLAB code.

- 1(c). Here are given two complex numbers: 11

$$Z_1 = 2 + 3.5i ; Z_2 = 1 + 1.5i$$

Use MATLAB code

- to find polar and exponential representation of Z_1 ,
- to find addition and multiplication of Z_1 and Z_2 .

- 2(a). For gases, the variation of viscosity with temperature is frequently modeled by the equation: 14

$$\mu = \frac{CT^{3/2}}{T + S}$$

Where, μ is the viscosity, T is temperature, C and S are empirical constant. Below is a table that gives viscosity of air at different temperatures.

$T (^{\circ}\text{C})$	-20	0	40	100	200	300	400	500
$\mu (\text{Ns/m}^2) (\times 10^{-5})$	1.63	1.71	1.87	2.17	2.53	2.98	3.32	3.64

Determine the constant C and S by curve-fitting the equation to the data points. Make a plot of viscosity versus temperature. In the plot show the data points with markers and curve-fitted equation with solid line. The curve-fitting can be done by rewriting the equation in the form $\frac{T^{3/2}}{\mu} = \frac{1}{C}T + \frac{S}{C}$ and using a first-order polynomial.

- 2(b). Consider the following two matrices: 09

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \text{ and } B = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}$$

By using MATLAB, perform the following tasks:

- Compute the matrix product AB and element-wise product of A and B .
- Find transpose, rank and inverse of A .
- Find determinant of B and scalar multiplication of B by 2.

2(c). Given that,

12

$$\frac{4x^2}{(x+1)^2(x-1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

Use a system of equations to find the partial fraction decomposition of the rational expression. Solve the system using MATLAB.

3(a). Consider the following system of linear equations:

12

$$3x_1 - 2x_2 + 5x_3 - 4x_4 + x_5 = 10$$

$$2x_1 + 4x_2 - 3x_3 + 6x_4 - 2x_5 = 20$$

$$5x_1 - 3x_2 + 2x_3 + x_4 + 4x_5 = 30$$

$$-x_1 + 6x_2 + x_3 - 2x_4 + 5x_5 = 40$$

$$4x_1 - 2x_2 + 3x_3 + 5x_4 - 6x_5 = 50$$

Write MATLAB code to

- Find the solution to this system by considering the augmental matrix of the system.
- Find the solution to this system by using the coefficient of the system using both left and right division.

3(b). Define LU and SVD decomposition and explain the physical significance of them. For the given matrix X , write MATLAB code to perform LU and SVD decomposition. Also construct the matrix from the decomposed matrices.

12

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

3(c). Define Eigen value and Eigen vectors with geometric interpretation of Eigen values. Using MATLAB commands, find the Eigen values and Eigen vectors of the following matrix C .

11

$$C = \begin{bmatrix} -1 & 0 & -2 \\ -1 & -3 & 0 \\ -6 & -2 & -1 \end{bmatrix}$$

4(a). The power generation of a PV system in a typical day is given in the following table:

13

Time (hour)	8:00	9:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00
Generation (kWh)	0.5	3.2	7.1	12.0	17.5	22.8	27.0	29.6	30.0	28.2
Radiation (kWh/m ²)	0.2	0.5	1.2	2.0	2.5	2.8	2.9	2.7	2.3	1.5

Plot time versus generation and time versus radiation using dual axis in MATLAB. Plot a red marker at the point where generation is maximum. Use proper axes labels, legends and keep the grid on.

4(b). The Van der Waals equation is given by, $P = \frac{nRT}{V-b} - \frac{n^2a}{V^2}$, where n (number of moles) are 1.5, R (gas constant) is 0.08206 ((L.atm)/(mol.k)), a is 1.39 L² atm/mol² and b is 0.03913 L/mol. Make a 3D plot that shows the variation of pressure (z axis) with volume (x axis) and temperature (y axis). The domain for the volume and temperature are $0.3 \leq V \leq 1.2$ L and $273 \leq T \leq 473$ K.

10

4(c). Write a user defined function that fits data points to a power function of the form $y = bx^m$. Name of the data points and the output arguments b and m are the constants of the fitted exponential equation. Use powerfit to fit the data below. Make a plot that shows the data points and the points with proper labelling.

12

$x =$	0.5	2.4	3.2	4.9	6.5	7.8
$y =$	0.8	9.3	37.9	68.2	155	198

SECTION – B

- 5(a). What is GNU Octave? What is GNU stand for? Write the application and limitation of GNU octave. Also, compare between octave and MATLAB. 09
- 5(b). Write a single line statement to solve the following problems in MATLAB: 12
- Generate a 5×6 matrix with random number ranging from 1 to 7 and assign the matrix to a variable.
 - Extract 1st, 3rd and 5th column from the generated matrix.
 - Delete the 2nd and 3rd row of the matrix.
 - Call the main diagonal of the matrix.
 - If A = [2 3 4 5] and B = [7 8 9]; what will be the output of [A, B] and [A ; B].
 - Check logically the elements that are greater than 4 in your generated matrix.
- 5(c). What is the purpose of using assignment operator in MATLAB? Explain it with example. Also, write down the rules that one must to follow during the naming of variable. 08
- 5(d). Write the output of the following matrix: 06

$$A = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 5 & 3 & 2 & 2 \end{bmatrix}$$

i. flip(A), ii. sort(A) iii. A(0;6) = 9

- 6(a). What is wrong with the following code (find at least three mistakes) 08
- S = input ("Enter a text string:");
If (S == "Hi");
 disp("your entered Hi");
elseif:
 disp("you did not enter Hi");
end
 - n = 200; a = 50
If n>50
 fprintf("you are wrong");
elseif n<50
 endif
end

- 6(b). Write MATLAB code that determines the volume of the water in the tank from the position (height h) of the float. The program must ask the user to enter a value of h in m, and as output displays the volume of the water in m³. 13

For $0 \leq h \leq 19$ m the volume of the water is given by the volume of a cylinder with height h: $V = \pi \times 12.5^2 h$. For $19 \leq h \leq 33$ m the volume of the water is given by adding the volume of a cylinder with $h = 19$ m, and the volume of the water in the cone:

$$V = \pi \times 12.5^2 \times 19 + \frac{1}{3} \pi (h - 19) (12.5^2 + 12.5 r_h + r_h^2)$$

$$\text{Where, } r_h = 12.5 + \frac{10.5}{14} (h - 19)$$

- 6(c). What are the output commands available in MATLAB? If current I=1.52 A and resistance R=20 Ω. Then write the MATLAB code that calculates the voltage drop V and show the formatted output as follows: 06
- "When current is 1.52 A and resistance is 20 Ω, the voltage drop is 30.4 volt."
- 6(d). Write short notes on the following function: (i) nthroot, (ii) ceil, (iii) all, (iv) rem 08
- 7(a). Write MATLAB code for solving the following series for calculating exponential value given by: 15

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Use for loop to calculate $e^{1.5}$ using first 20 terms of the series above.

7(b). Write down the output of the following codes:

12

```
i) n = 5;
   for k = 0:0.5:n
       disp("ESE 2K22")
       if k>3.5
           break
       end
   end
```

```
ii) m = 1
    While 1
        disp("ESE-New")
        m = m + 1;
        if m> 10
            break
        end
    end
```

7(c). Beta = [3, 7, 12, 4, 5, 8, 9, 11, 15, 14];

08

```
Countd = 0;
For i=1: length(Beta)
    if rem(Beta(i),2) == 0
        Continue;
    end
    Countd = Countd+1;
end
disp(countd)
```

Explain each line of this code.

8(a). Define i. Local variable, ii. Global variable, iii. Function. Also write how "H1 line" and "help text lines" work in a function file.

08

8(b). Define local and global variables with MATLAB examples. Write an anonymous MATLAB function for the following math function:

13

$$y(x) = 0.6x^3e^{-0.47x} + 1.5x^2e^{-0.6x}$$

The input to the function is x and the output is y . Write the function such that x can be a vector (use element by element operations)

- Use the function to calculate $y(-2)$ and $y(4)$.
- Use the function to calculate $y(x)$ for $-4 \leq x \leq 8$.

8(c). Write a user-defined MATLAB function for the following math function:

14

$$r(\theta) = 3 \sin(3 \cos(0.5\theta))$$

The input to the function is θ (in radians) and the output is r . Write the function such that θ can be vector.

- Use the function to calculate $r(\frac{\pi}{6})$ and $r(\frac{5\pi}{6})$.
- Use the function to plot polar plot $r(\theta)$ for $0 \leq \theta \leq 4\pi$

--- End ---